

REDDING, CA, USA

WMO: 725920

Lat: 40.518N

Lon: 122.299W

Elev: 497

StdP: 14.43

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24257

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.4	31.8	11.1	9.9	55.2	15.5	12.3	55.7	34.1	51.7	28.7	52.8	1.2	30	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.0	105.4	68.0	102.4	67.1	99.4	66.0	71.4	96.5	69.9	94.5	68.5	92.7	7.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	89.4	79.4	61.4	82.8	78.3	59.5	77.3	77.0	35.4	97.3	34.1	94.1	32.9	93.0	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	20.0	16.8	DB	24.5	111.2	3.2	2.7	22.1	113.1	20.2	114.6	18.4	116.1	16.1	118.0	(4)
(5)			WB	22.9	74.1	3.1	1.6	20.7	75.2	18.8	76.2	17.1	77.1	14.8	78.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.2	47.4	50.1	54.4	59.1	67.7	76.8	83.0	80.6	74.9	64.2	52.7	46.6		
	DBStd	14.26	5.59	6.04	6.55	7.40	7.80	7.83	5.32	4.92	6.36	6.66	6.42	5.56		
	HDD50	399	116	67	29	9	0	0	0	0	0	0	43	135		
	HDD65	2619	545	419	334	207	63	8	0	0	6	96	370	571		
	CDD50	5218	37	68	165	282	550	804	1023	948	747	439	126	29		
	CDD65	1961	0	1	6	30	147	362	558	483	302	70	2	0		
	CDH74	27663	6	17	124	505	1980	4967	8084	6758	4079	1074	68	1		
	CDH80	15939	0	1	19	161	904	2852	5123	4161	2298	409	11	0		
(14) Wind	WSAvg	5.8	5.2	6.6	6.6	6.5	6.5	6.7	5.6	5.1	5.2	5.5	4.6	5.4		
(15) Precipitation	PrecAvg	37.9	7.0	5.7	5.2	2.5	1.6	0.7	0.1	0.4	0.9	2.0	5.2	6.6		
	PrecMax	73.1	27.0	15.8	16.4	9.2	7.7	2.6	3.3	5.5	7.4	8.8	16.2	17.7		
	PrecMin	20.1	0.3	0.2	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0		
	PrecStd	11.1	5.3	4.1	3.7	2.0	1.9	0.7	0.5	0.9	1.5	1.9	3.9	4.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.2	76.8	82.7	89.7	98.7	108.1	110.3	108.4	104.3	95.2	81.1	69.3		
		MCWB	52.2	55.9	58.2	61.0	65.2	68.5	69.5	69.1	65.4	62.4	57.7	50.4		
	2%	DB	67.9	71.1	77.5	84.8	93.7	102.6	106.0	103.9	99.9	88.8	75.0	63.9		
		MCWB	50.1	52.9	56.9	59.8	64.4	67.2	68.7	67.2	64.1	60.8	56.1	48.9		
	5%	DB	62.2	66.3	73.0	80.2	89.6	98.4	102.9	100.5	96.5	84.5	70.2	59.8		
		MCWB	48.9	51.1	55.0	58.6	62.6	65.7	67.9	66.2	63.6	58.9	54.2	48.7		
	10%	DB	57.8	61.6	68.3	75.0	85.0	94.3	99.7	97.5	93.0	80.1	65.1	56.4		
		MCWB	48.0	49.7	53.0	56.6	61.0	64.4	66.9	65.3	63.0	57.5	52.0	48.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	58.3	60.9	64.9	69.2	72.7	73.6	72.6	71.1	64.9	61.2	57.1		
		MCDB	62.2	68.0	75.3	82.4	89.0	95.9	99.0	100.5	91.0	87.0	74.6	59.4		
	2%	WB	53.7	56.3	58.8	62.1	66.5	70.0	71.8	70.3	67.8	62.9	58.6	54.5		
		MCDB	59.2	64.9	72.4	79.4	87.6	93.6	98.1	95.3	91.1	83.4	69.4	57.2		
	5%	WB	52.0	54.0	56.9	59.9	64.6	68.1	70.4	68.6	65.9	61.1	56.5	52.6		
		MCDB	57.4	62.0	68.7	76.1	85.1	92.2	96.1	93.2	89.6	79.3	65.3	55.6		
	10%	WB	50.6	51.9	54.8	57.9	62.7	66.3	69.0	67.2	64.2	59.3	54.7	50.5		
		MCDB	56.3	58.8	65.3	72.1	81.4	89.6	93.9	91.4	88.4	75.3	62.0	54.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	18.6	20.7	22.9	25.3	27.6	31.0	32.2	31.1	26.7	20.8	16.7		
		MCDBR	29.7	28.5	29.5	31.5	32.6	32.5	34.7	35.9	37.0	34.6	30.6	25.2		
		MCWBR	15.6	14.5	13.8	13.0	10.8	9.6	9.5	10.3	11.8	13.4	14.9	14.3		
		MCDBR	19.7	21.8	25.4	28.7	29.8	30.8	31.9	32.5	32.4	30.0	24.3	13.9		
	5% WB	MCWBR	11.8	12.2	12.9	12.7	10.8	9.9	9.4	10.0	11.1	12.4	12.8	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.319	0.329	0.347	0.351	0.350	0.354	0.341	0.339	0.329	0.309	0.314		
	taud		2.535	2.524	2.508	2.443	2.471	2.488	2.473	2.500	2.494	2.533	2.566	2.528		
	Ebn at Noon		271	286	294	294	294	292	291	292	286	278	271	260		
	Edh at Noon		23	27	30	35	35	34	34	32	30	26	22	22		
(44) All-Sky Solar Radiation	RadAvg		642	927	1313	1803	2233	2555	2599	2309	1840	1242	751	539		
	RadStd		122	134	174	186	165	145	83	82	76	113	86	113		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.92	N/A	-2.66	N/A	N/A	N/A	-67	-167	+299
(47) Regional (16 neighbors)		N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	+184

Nomenclature: See separate page